DAMIAN MARQUEZ

Junior Backend Engineer | Python & FastAPI | AI/LLM Integrations

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SUMMARY

Junior backend engineer working with Python, FastAPI, PostgreSQL and LLM integrations. I designed, built and deployed **Repuesthogar**, a product catalog platform for a retail client in Uruguay with 758 products live: FastAPI API, PostgreSQL on Neon, JWT-protected admin and Cloudinary image storage. My personal project **ExamGen AI** is a FastAPI backend that generates complete exams with an LLM using structured JSON output, with user-scoped authorization and PDF export. Looking to join a remote backend team.

TECHNICAL SKILLS

Backend & APIs	Python, FastAPI, REST API design, Pydantic, SQLAlchemy, OpenAPI documentation
Databases	PostgreSQL, SQL, data modeling, Alembic migrations, Neon
Security	JWT authentication, password hashing (bcrypt), authorization with user-scoped queries, input validation
AI & LLM	OpenAI API and OpenAI-compatible APIs (Groq), structured JSON output, prompt design, embeddings
Infrastructure & tools	Docker, Docker Compose, Nginx, Git/GitHub, Linux, Render, Cloudflare, Cloudinary, ReportLab
Frontend (secondary)	React, TypeScript, JavaScript, Vite, Next.js, Tailwind CSS

EXPERIENCE

Freelance Developer

Independent · Uruguay

Repuesthogar — Product Catalog Platform (client project) repuesthogar.com

Python · FastAPI · PostgreSQL (Neon) · SQLAlchemy · Alembic · JWT · Cloudinary · React · TypeScript · Render · Cloudflare Pages

- Built and deployed a product catalog for an appliance and spare-parts store: 758 products in 12 categories, each with code, description and photo. Customer inquiries go to WhatsApp with the product name pre-filled.
- Designed a FastAPI REST API that serves the public catalog read-only and exposes JWT-protected admin endpoints for product CRUD and image upload.
- Modeled the schema with SQLAlchemy and managed every change through Alembic migrations on PostgreSQL (Neon).
- Normalized supplier product lists into one consistent dataset and imported it into the database.
- Offloaded image storage and delivery to Cloudinary; the API stores only image URLs.
- Deployed the frontend on Cloudflare Pages, the API on Render and the database on Neon; moved DNS to Cloudflare for the client's domain.

Landing website (client project)

- Delivered a business landing website for a second local client, from requirements to production.

PROJECTS

ExamGen AI — AI-Powered Exam Generation (personal project) github.com/DevDamianMarquez/examgen_ai

Python · FastAPI · Pydantic · SQLAlchemy · PostgreSQL · JWT · OpenAI SDK (Groq) · ReportLab · Docker Compose · React

- Built a FastAPI backend that turns a syllabus into a complete exam (questions, answer key, grading rubric) using an LLM in JSON mode; 15 REST endpoints documented through OpenAPI.
- Implemented JWT authentication with bcrypt password hashing (SHA-256 pre-hash to avoid bcrypt's 72-byte limit) and user-scoped queries so users can only access their own data; exam generation requires an active subscription.
- Modeled 6 tables (users, subjects, syllabuses, exams, questions, subscriptions) and stored the raw LLM JSON alongside normalized question rows.
- Reduced repeated questions by feeding questions from previous exams back into the prompt; built embedding and cosine-similarity utilities for semantic duplicate detection (in progress).
- Generated student and teacher PDF versions on demand with ReportLab.
- Containerized with Docker and Docker Compose (PostgreSQL 16 with health check) and wrote an Nginx config to serve the React client and proxy the API.

EDUCATION & LANGUAGES

Full-Stack Development Program, BIOS EDU · 10-month program · Completed December 2024

Spanish: Native · **English:** intermediate reading and writing, basic listening and speaking